

Online Appendix

The Macroeconomic Effects of Global Supply Chain Shocks

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April 28, 2026

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A Sensitivity Checks

This appendix presents the results of a series of robustness checks on both the properties of the instrument and the stability of the main findings, including supplementary checks in addition to the tests discussed in the main text.

A.1 Diagnostics of the Instrument

To assess the validity of the instrument, I conduct a set of diagnostic checks following the approach outlined in Ramey (2016). Figure A.1 plots the autocorrelation function of the FEU price surcharge series. The autocorrelations are statistically indistinguishable from zero at all lags, and the Ljung–Box Q-statistic testing the joint null hypothesis that all autocorrelations up to lag 20 are zero yields a p-value of 0.39. This provides no evidence of serial correlation.

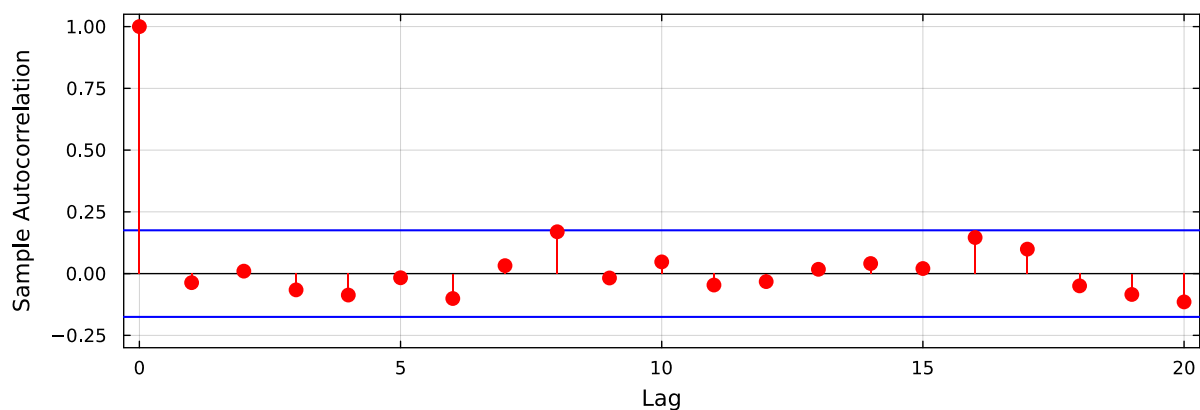


Figure A.1: The Autocorrelation Function of the FEU Price Surcharge Series

Notes: The figure shows the autocorrelation function. The p-value for the Ljung–Box Q-statistic testing whether all autocorrelations up to lag 20 are jointly zero is 0.39.

I also examine whether the FEU price surcharge series can be forecasted using a broad set of macroeconomic variables. Table A.1 reports the results of a series of Granger causality tests where the FEU price surcharge series is regressed on its own twelve lags and two lags of each control variable. The p-values are all well above conventional significance thresholds,

Table A.1: Granger Causality Tests

Series	Source	Transf	p-value	F-stat	df1	df2	Sample
World Oil Production	Baumeister and Hamilton (2019)	MoM	0.76	0.27	2	98	2014M8-2024M12
US Consumer Price Index	Haver Analytics	MoM	0.56	0.58	2	98	2014M8-2024M12
US Terms of Trade	Haver Analytics	Level	0.32	1.16	2	98	2014M8-2024M12
WTI Oil Price	Haver Analytics	MoM	0.72	0.33	2	98	2014M8-2024M12
US Industrial Production	Haver Analytics	MoM	0.74	0.30	2	98	2014M8-2024M12
US Unemployment Rate	Haver Analytics	Level	0.53	0.63	2	98	2014M8-2024M12
GSCPI	Benigno et al. (2022)	Level	0.43	0.85	2	98	2014M8-2024M12
World Uncertainty Index	Ahir, Bloom and Furceri (2022)	Level	0.79	0.24	2	98	2014M8-2024M12
US Trade Policy Uncertainty Index	Baker, Bloom and Davis (2016)	Level	0.43	0.86	2	98	2014M8-2024M12
World Sentiment Index	Ahir, Bloom and Furceri (2022)	Level	0.54	0.62	2	98	2014M8-2024M12
S&P500	Haver Analytics	MoM	0.66	0.42	2	98	2014M8-2024M12
Geopolitical Risk Index	Caldara and Iacoviello (2022)	Level	0.78	0.25	2	98	2014M8-2024M12
Global Economic Condition	Baumeister, Korobilis and Lee (2022)	Level	0.78	0.25	2	98	2014M8-2024M12
Oil Price Expectations 3Months	Baumeister (2023)	Level	0.77	0.26	2	78	2014M8-2023M4
US - China Tension Index	Rogers, Sun and Sun (2024)	Level	0.61	0.50	2	89	2014M8-2024M3
Federal Funds Rate	FRED	Level	0.62	0.47	2	98	2014M8-2024M12

Notes: The table shows the results of a series of Granger Causality tests of the FEU price surcharge series using a selection of macroeconomic variables. The series are made stationary when necessary by taking the MoM growth rate. The tests are conducted by regressing the shock on its own 12 lags, 2 lags of the other variable and a constant. The test is on the joint significance of the lags of the additional variable. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

and the null hypothesis that the instrument is not Granger-caused by other variables cannot be rejected for any series. These results suggest that the instrument is not systematically predictable based on past macroeconomic indicators or indices capturing global uncertainty, geopolitical risk, and US–China trade conditions.

Next, I assess whether the instrument is correlated with structural shock series commonly used in the literature. Table A.2 reports the Pearson correlation coefficients between the FEU price surcharge series and a wide range of structural shocks, including oil demand and supply shocks, monetary policy shocks, and other shocks. None of the correlations is statistically significant. Taken together, these diagnostic results support the interpretation of the FEU price surcharge series as a valid external instrument. The series is serially uncorrelated, not forecastable from past macroeconomic variables, and orthogonal to other structural shocks commonly used in the literature, reinforcing the exogeneity condition required for Proxy-SVAR identification.

Table A.2: Correlation with other Structural Shocks

Series	Source	ρ	p-value	t-stat	df	Sample
Oil News Shock	Känzig (2021)	-0.05	0.62	-0.50	123	2014M8-2024M12
Carbon News Shock	Känzig (2023)	0.11	0.37	0.91	63	2014M8-2019M12
Oil Supply Shock	Baumeister and Hamilton (2019)	-0.03	0.72	-0.36	123	2014M8-2024M12
Economic Activity Shock	Baumeister and Hamilton (2019)	0.02	0.86	0.18	123	2014M8-2024M12
Oil Consumption Demand Shock	Baumeister and Hamilton (2019)	-0.14	0.12	-1.56	123	2014M8-2024M12
Oil Inventory Demand Shock	Baumeister and Hamilton (2019)	0.01	0.91	0.12	123	2014M8-2024M12
Monetary Policy Shock	Bauer and Swanson (2023)	-0.05	0.61	-0.51	111	2014M8-2023M12

Notes: The table shows the correlation of the FEU price surcharge series with a wide range of structural shocks from the literature. ρ is the Pearson correlation coefficient, the p-value corresponds to the test whether the correlation is different from zero, t-stat is the corresponding test statistic, df are the degrees of freedom. * p<0.05, ** p<0.01, *** p<0.001.

A.2 Robustness of Main Results

To ensure the reliability of the baseline results, I perform a comprehensive set of robustness checks, including tests of the invertibility assumption, lag length sensitivity, and the inclusion of additional control variables. First, I test the invertibility assumption following the procedure outlined by Plagborg-Møller and Wolf (2022) and Hamilton (unpublished results). The first panel of Table A.3 reports the p-values from a set of F-tests conducted for each equation in the baseline VAR model, for increasing lag lengths of the instrument. Across all series and lag specifications, the p-values are above conventional significance levels, indicating that the coefficients on the lagged instrument are statistically indistinguishable from zero. Moreover, the joint test of the null hypothesis that all coefficients on past realizations of the instrument are jointly zero, reported in the second panel of Table A.3, cannot be rejected. Taken together, these results support the validity of the invertibility assumption and suggest that the baseline VAR does not suffer from omitted information problems.

Second, I assess the sensitivity of the aggregate IRFs to alternative lag lengths. Figure A.2 displays structural impulse responses estimated from VAR specifications including 14, 16, 18, and 20 lags, in addition to the 12-lag baseline. The results are robust across specifications, with the dynamic causal effects of the global supply chain shock remaining statistically

Table A.3: Testing the invertibility assumption

Lags	US CPI	WTI Oil Price	US Ind Prod	US Unemployment	GSCPI
6	0.11	0.26	0.96	0.82	0.68
7	0.14	0.10	0.97	0.87	0.79
8	0.14	0.11	0.99	0.90	0.68
9	0.15	0.12	0.98	0.92	0.44
10	0.21	0.15	0.98	0.95	0.30
11	0.28	0.19	0.97	0.97	0.33
12	0.25	0.18	0.98	0.98	0.42
Joint test: 12 lags p-value = 0.99 F-stat = 0.58 $df_1 = 60$ $df_2 = 200$					

Note: The first panel shows the p-values of a series of F-test that the coefficients $\alpha_1, \dots, \alpha_m$ are zero in the regression $y_{it} = \pi_i' \mathbf{x}_{t-1} + \alpha_1 z_{t-1} + \dots + \alpha_m z_{t-m} + \eta_{it}$. The test is conducted for each series included in the baseline VAR model, for different numbers of lags of the instrument. The lag order of the VAR is set to 12 and in terms of deterministics, only a constant is included. The second panel shows the joint system test across all i that lags of z do not appear in any of the equations. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

and economically consistent regardless of lag length. One exception is the WTI oil price series, which exhibits notable variation across specifications; however, these responses are not statistically significant and have wide confidence intervals. The main takeaway is that the results are stable across lag structures and consistent with the baseline specification.

Lastly, I examine whether the impulse responses are robust to the inclusion of additional control variables. Specifically, I estimate alternative specifications of the baseline VAR model by sequentially adding one control variable at a time from a broad set of economic indicators, including price expectations, sectoral price and production indices, services, world output, interest rates, and other macro-financial variables (see Appendix B for details).

Figure A.3 shows that the estimated impulse responses remain remarkably stable across these specifications. These results demonstrate that the dynamic causal effects of global supply chain shocks are robust not only to lag structure but also to the inclusion of a wide range of controls. This reinforces the credibility of the baseline findings.

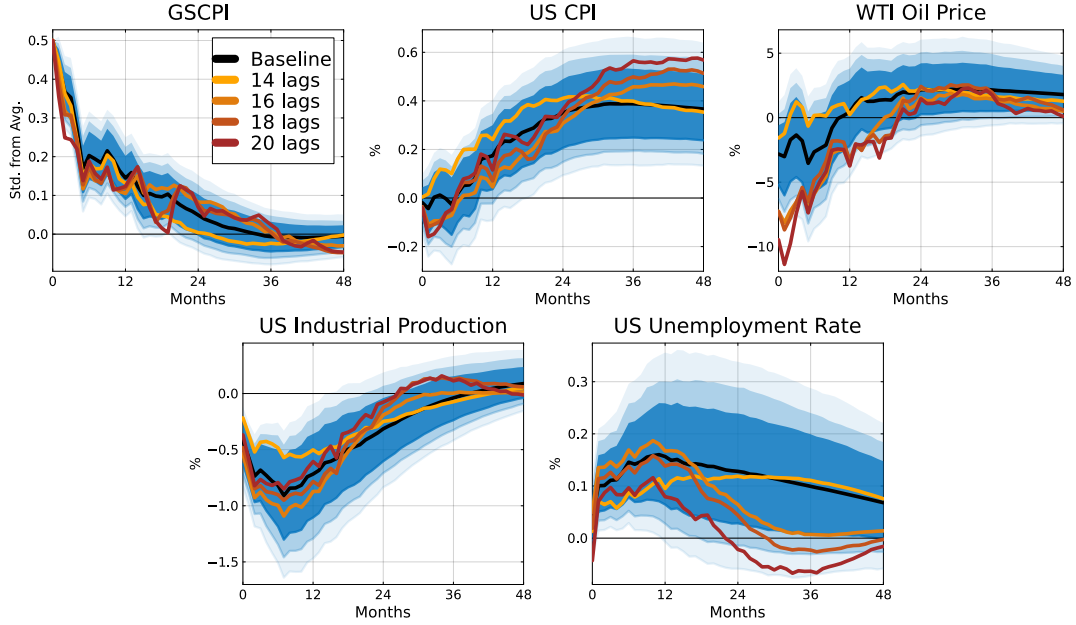


Figure A.2: Robustness with respect to Lag Length

Notes: Impulse responses to a global supply chain shock corresponding to a \$1,340 price surcharge. The solid lines are the point estimates and the shaded areas are 68, 80 and 90 percent confidence bands for the baseline model, respectively.

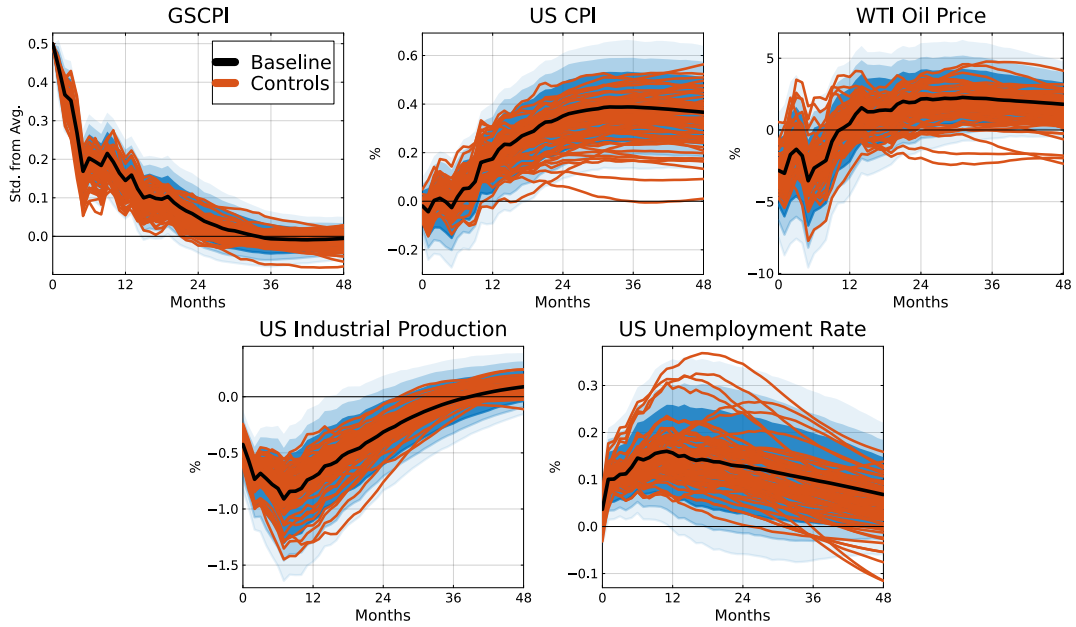


Figure A.3: Robustness with respect to Additional Controls

Notes: Impulse responses to a global supply chain shock corresponding to a \$1,340 price surcharge. The solid black line is the point estimate and the shaded areas are 68, 80 and 90 percent confidence bands for the baseline model. The solid orange lines are the point estimates for the model with additional controls.

B Data

This appendix provides details on the variables used in the aggregate and sectoral analysis, as well as those included in the robustness check involving additional control variables. Specifically, Table B.4 lists the series ID, series name, data source, and the transformation applied to each variable. The table is divided into panels, each containing the variables used in a specific analysis, as indicated by the corresponding panel title. All variables in Panels B through E were added individually to the baseline specification to test the robustness of the main results to additional controls (Figure A.3). Variables used exclusively in the Granger causality tests (Table A.1) or the correlation analysis with other structural shocks (Table A.2) are not included, as their sources are already documented in the corresponding tables. All variables were used over the sample period from January 1998 to December 2024.

Table B.4: Data description

Series ID	Series Name	Data Source	Transformation
Panel A. Aggregate Analysis			
GSCPI@NY	Global Supply Chain Pressure Index	New York FED	Level
S111PC@G10	US CPI	Haver Analytics	100 log
N111POIL@G10	WTI Oil Price	Haver Analytics	100 log
S111D@G10	US Industrial Production	Haver Analytics	100 log
S111ELUR@G10	US Unemployment Rate	Haver Analytics	Level
Panel B. Sectoral Production Analysis			
IPMFG@USECON	US IP Manufacturing	Haver Analytics	100 log
IPB0@USECON	US IP Mining	Haver Analytics	100 log
IPMDG@USECON	US IP Durable Goods	Haver Analytics	100 log
IPMND@USECON	US IP Nondurable Goods	Haver Analytics	100 log
IPE1T2@USECON	US IP Food and Beverages	Haver Analytics	100 log
IPG61T3@USECON	US IP Motor Vehicles and Parts	Haver Analytics	100 log
IP53@USECON	US IP Materials	Haver Analytics	100 log
IPFP@USECON	US IP Final Products	Haver Analytics	100 log
Panel C. Non-Tradable Analysis			
NMFBAIA@USECON	Services Activity Index	Haver Analytics	100 log
NMFEIA@USECON	Services Employment Index	Haver Analytics	100 log
NAPMOI@USECON	Manufacturing Production Index	Haver Analytics	100 log
Panel D. Sectoral Prices Analysis			
CPILFESL	US CPI Core	FRED	100 log
PCUCS@USECON	US CPI Services	Haver Analytics	100 log
PCUCCD@USECON	US CPI Durable Goods	Haver Analytics	100 log
PCUSND@USECON	US CPI Nondurable Goods	Haver Analytics	100 log
S111PCFB@G10	US CPI Food and Beverages	Haver Analytics	100 log
PCUTPV@USECON	US CPI New Vehicles	Haver Analytics	100 log
S111PCH@G10	US CPI Housing	Haver Analytics	100 log
S111PCM@G10	US CPI Medical Care	Haver Analytics	100 log
Panel E. Additional Controls			
H111PFTT@G10	US Terms of Trade	Haver Analytics	100 log
FFR	Fed Funds Rate	FRED	Level
CINF5@USECON	Inflation Expectations 5Y	Haver Analytics	Level
CINF1@USECON	Inflation Expectations 1Y	Haver Analytics	Level
S001DPP@G10	World Industrial Production	Haver Analytics	100 log

CUSR0000SEHF	US CPI Energy Services	FRED	100 log
PCUCC@USECON	US CPI Commodities	Haver Analytics	100 log
S111PCH@G10	US CPI Recreation	Haver Analytics	100 log
PCUHF@USECON	US CPI Fuels and Utilities	Haver Analytics	100 log
S111PCT@G10	US CPI Transportation	Haver Analytics	100 log
S111PCO@G10	US CPI Others	Haver Analytics	100 log
PCUSE@USECON	US CPI Energy	Haver Analytics	100 log
PCUTV@USECON	US CPI Motor Vehicles	Haver Analytics	100 log
PCUA@USECON	US CPI Apparel	Haver Analytics	100 log
UHFGE@USECON	US CPI Household Electricity	Haver Analytics	100 log
UHFGU@USECON	US CPI Piped Gas Service	Haver Analytics	100 log
PPIACO	US PPI	FRED	100 log
PCUOMFGOMFG	US PPI Manufacturing	FRED	100 log
PCU311311	US PPI Food Manufacturing	FRED	100 log
IP521@USECON	US IP Business Equipment	Haver Analytics	100 log
IPHT@USECON	US IP High Tech Industries	Haver Analytics	100 log
IP54@USECON	US IP Nonindustrial Supplies	Haver Analytics	100 log
IPUTL@USECON	US IP Electric and Gas Utilities	Haver Analytics	100 log
IPG3@USECON	US IP Machinery	Haver Analytics	100 log
IPG61@USECON	US IP Motor Vehicles	Haver Analytics	100 log
IPF1@USECON	US IP Wood	Haver Analytics	100 log
IPF7@USECON	US IP Nonmetallic Mineral Products	Haver Analytics	100 log
IPG1@USECON	US IP Primary Metals	Haver Analytics	100 log
IPG2@USECON	US IP Fabricated Metal Products	Haver Analytics	100 log
IPG4@USECON	US IP Comp and Elec Comp	Haver Analytics	100 log
IPG5@USECON	US IP Elec Eqpt, Appliances	Haver Analytics	100 log
IPG64T9@USECON	US IP Aerospace	Haver Analytics	100 log
IPG7@USECON	US IP Furniture	Haver Analytics	100 log
IPE1T2@USECON	US IP Food Bev and Tob	Haver Analytics	100 log
IPE3T4@USECON	US IP Textile	Haver Analytics	100 log
IPE5T6@USECON	US IP Apparel	Haver Analytics	100 log
IPF2@USECON	US IP Paper	Haver Analytics	100 log
IPF3@USECON	US IP Printing and Rel	Haver Analytics	100 log
IPF5@USECON	US IP Chemicals	Haver Analytics	100 log
IPF4@USECON	US IP Petroleum and Coal	Haver Analytics	100 log
IPF6@USECON	US IP Plastics and Rubber	Haver Analytics	100 log
IPN221113S	US IP Nuclear El Gen	FRED	100 log
IPB51222S	US IP Residential En Utilities	FRED	100 log

IPG2211S	US IP Electric Power	FRED	100 log
IPG2212S	US IP Gas Distribution	FRED	100 log
NMFVDI@USECON	Services Supplier Deliveries Index	Haver Analytics	100 log
NRSV2@USECON	Food Drinking Services	Haver Analytics	100 log
	Real Food Services (CPI Defl.)	Own Calculation	100 log
	Real Food Services (CPI Food Defl.)	Own Calculation	100 log
NAPMC@USECON	Manufacturing Composite Index	Haver Analytics	100 log
NAPMEI@USECON	Manufacturing Employment Index	Haver Analytics	100 log

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